

I went looking for a Quantum Advantage in Text Security.

Here is what the empirical data actually revealed.

01 | CONTROLLED BENCHMARK

Fidelity kernels vs RBF & Linear across 10 matched seeds & 2D-12D.

02 | THE 'VANISHING' EFFECT

Why an initial +1.14 pp MiniLM gain collapsed to -2.23 pp upon replication.

03 | CIRCUIT DEPTH PARADOX

Why adding ZZFeatureMap layers degraded F1 from 0.533 down to 0.320.

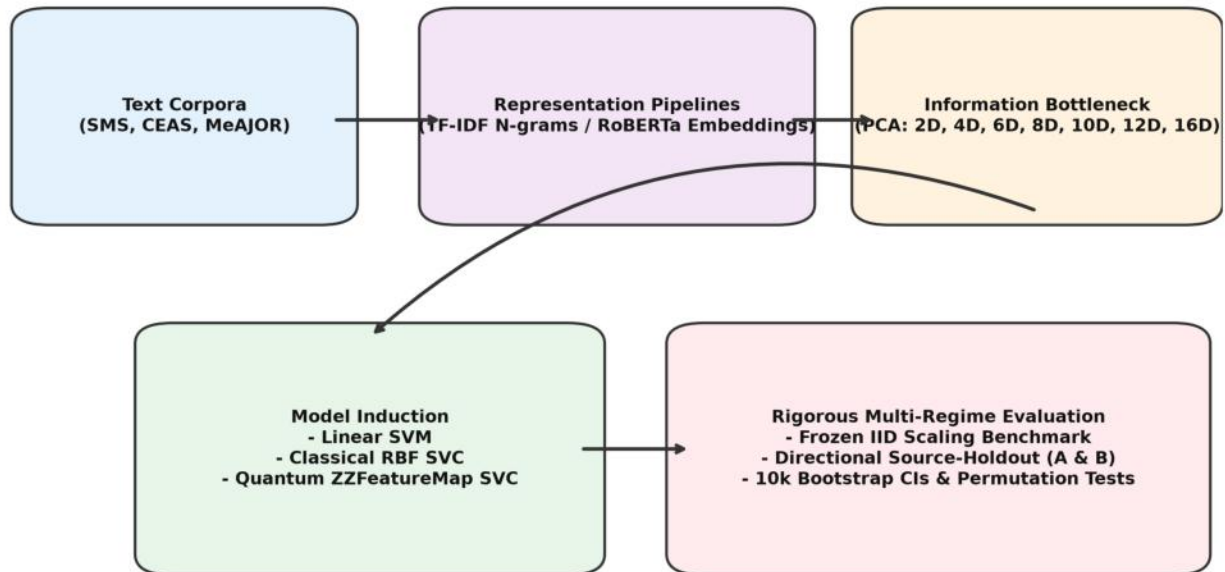
04 | GEOMETRIC MECHANISM

Why classical distance preservation ($r = 0.77$) governs quantum F1.

Controlled Evaluation Pipeline

- Compared TF-IDF against Sentence Transformers (RoBERTa, MiniLM, MPNet).
- Evaluated Quantum Fidelity Kernels (cyclic ZZFeatureMap) against RBF & Linear SVM.
- Benchmarked across 10 random seeds, 2D-12D dimensions, and domain shift.

Figure 1: Methodological Architecture of the Quantum vs Classical NLP Study



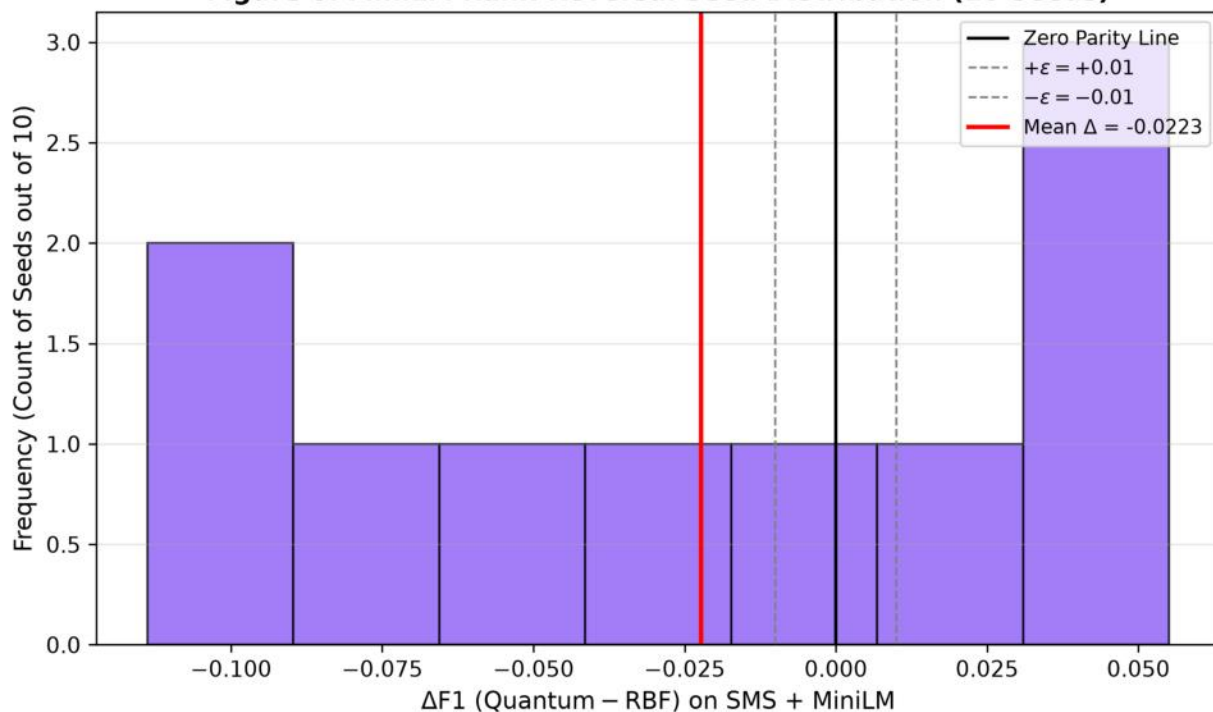
KEY TAKEAWAY

Evaluating across matched seeds and dimensions prevents cherry-picked isolated metrics.

The 'Vanishing' Advantage: 3 Seeds vs 10 Seeds

- Initial 3-seed run on MiniLM showed an apparent +1.14 pp F1 quantum advantage.
- Under rigorous 10-seed paired replication, confirmed difference was -2.23 pp ($p > 0.05$).
- Initial perceived gain was an artifact of small-sample seed variance.

Figure 6: MiniLM Rank-Reversal Seed Distribution (10 Seeds)

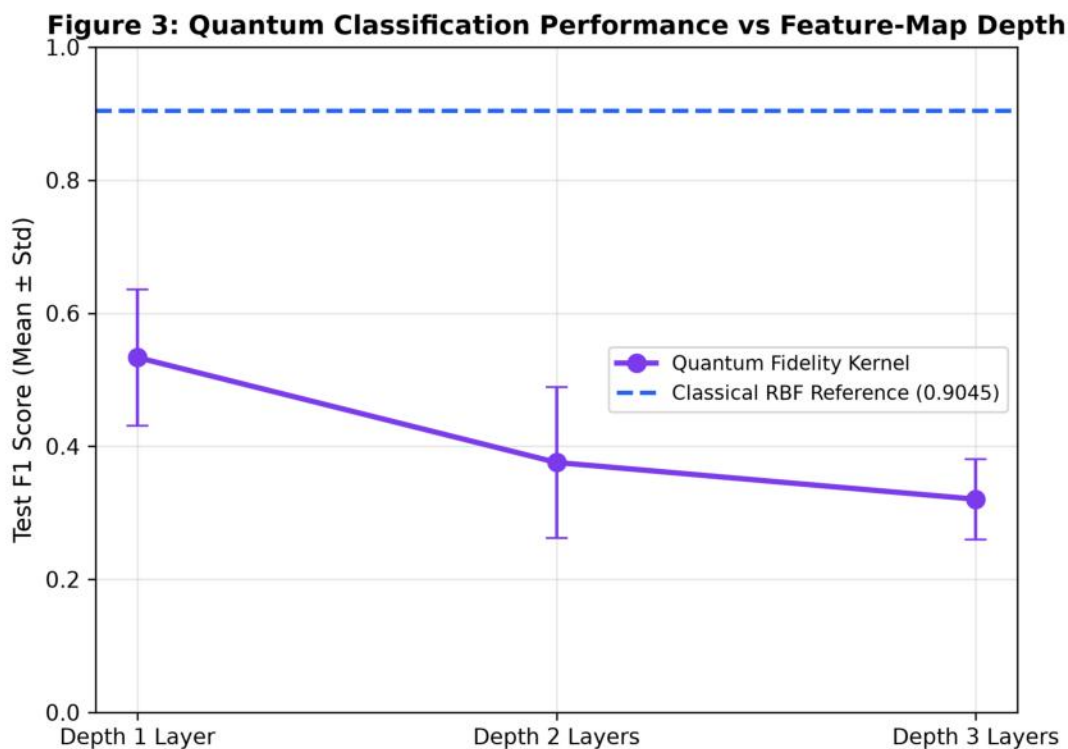


KEY TAKEAWAY

Finding a positive result is not the same as finding a statistically reliable result.

Circuit Depth Paradox: Deeper Maps Degrade F1

- Ablated cyclic ZZFeatureMap depth: 1 Layer (0.533 F1) → 2 Layers (0.375) → 3 Layers (0.320).
- Deeper quantum circuits did not enhance representation power on sentence embeddings.
- Instead, unparameterized entanglement layers caused severe orthogonal state collapse.

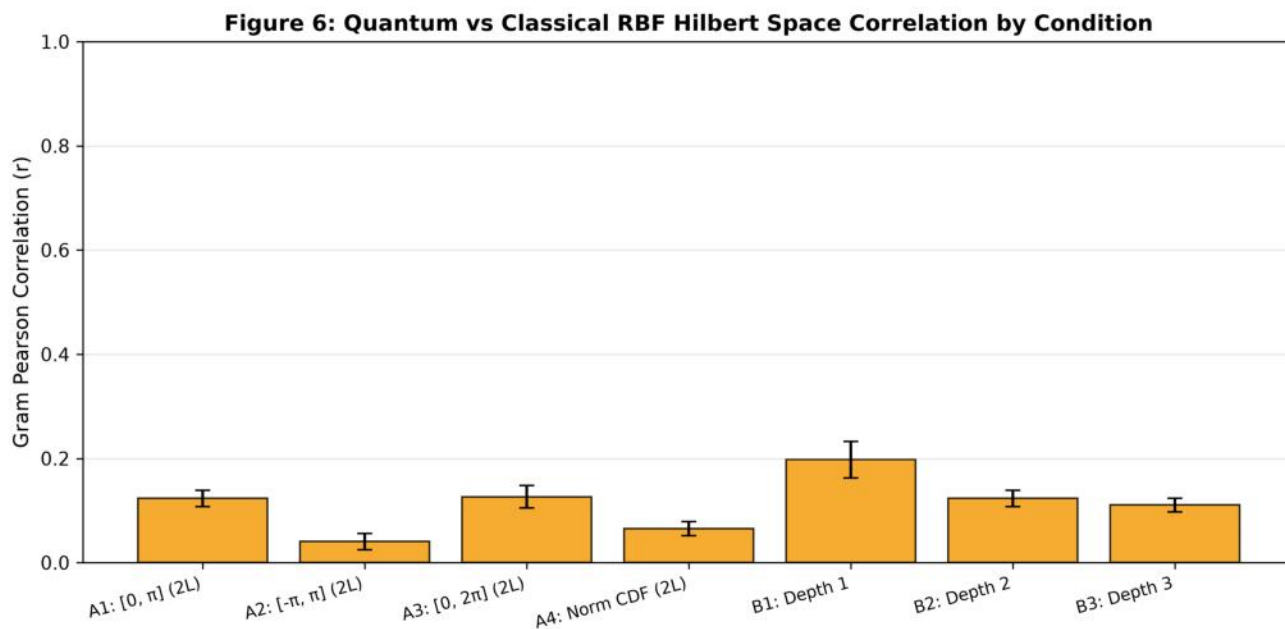


KEY TAKEAWAY

Deeper quantum circuits without trainable parameters can distort metric spaces.

Distance Preservation ($r = 0.77$) Explains Performance

- Quantum F1 correlates strongly with classical distance preservation ($r = 0.77$).
- Kernel diversity alone shows virtually zero correlation with F1 ($r = -0.05$).
- Core question: When is text embedding geometry compatible with a quantum feature map?

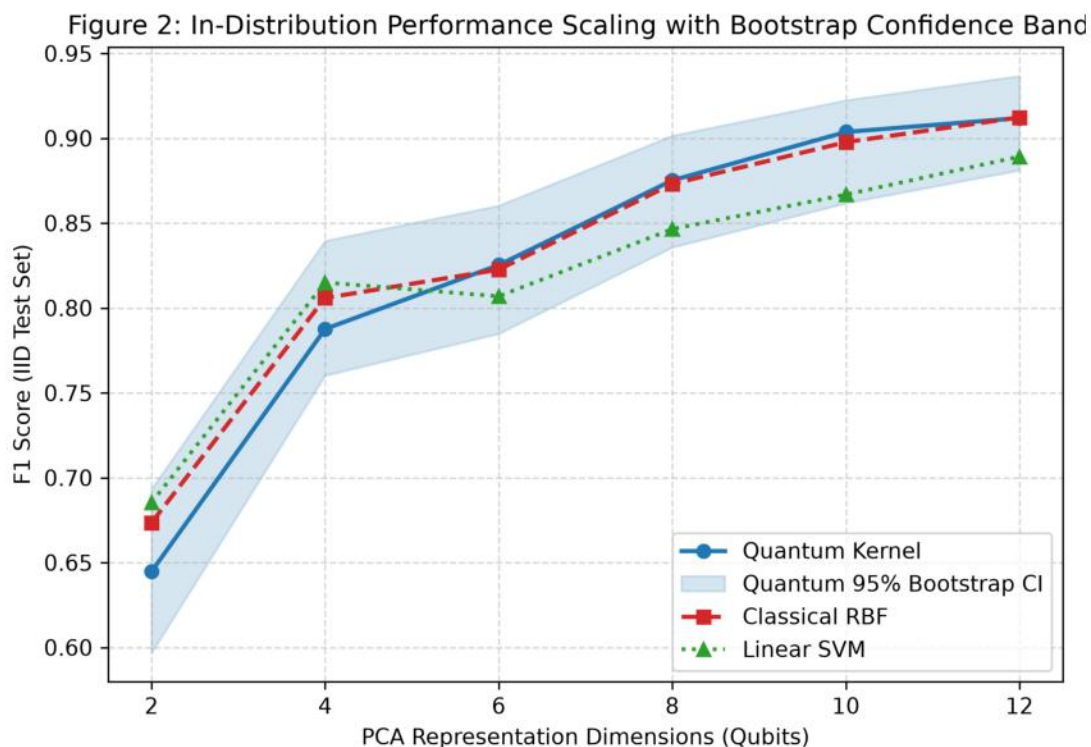


KEY TAKEAWAY

Quantum kernels succeed when they preserve meaningful metric structure, not raw diversity.

Key Findings & Interactive Dashboard

- Quantum Fidelity Kernels matched classical RBF under IID, but degraded under source shift.
- 12D simulation incurred a $\sim 64\times$ computational penalty without a net F1 gain.
- Explore the live open-source research dashboard: quantum-three-hazel.vercel.app



KEY TAKEAWAY

A rigorous negative result provides more actionable scientific insight than a false positive.